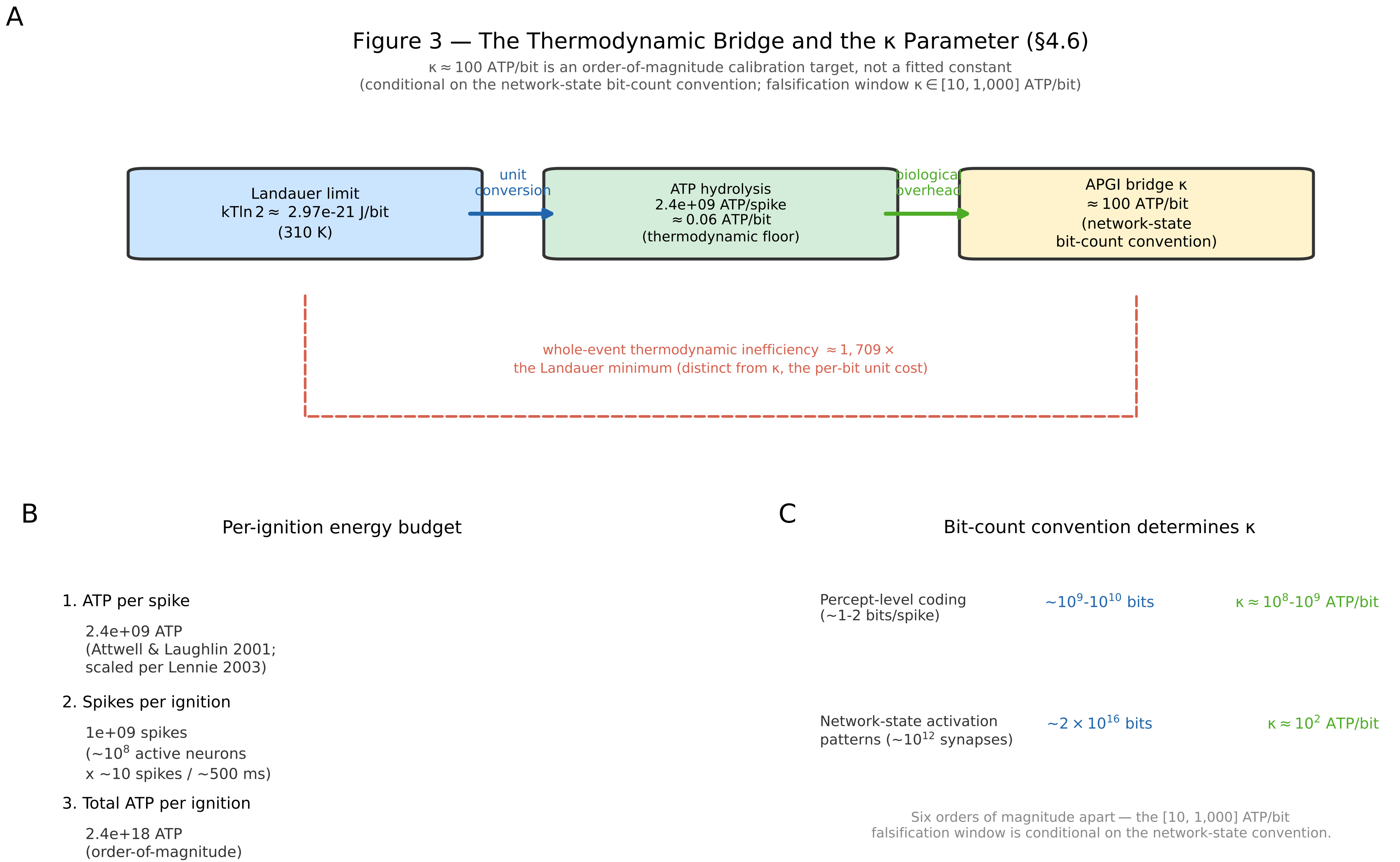
A
Figure 3 — The Thermodynamic Bridge and the κ Parameter (§4.6)
κ ≈ 100 ATP/bit is an order-of-magnitude calibration target, not a fitted constant
(conditional on the network-state bit-count convention; falsification window κ ∈ [10, 1,000] ATP/bit)
Landauer limit
kTln 2 ≈ 2.97e-21 J/bit
(310 K)
unit conversion
ATP hydrolysis
2.4e+09 ATP/spike
≈ 0.06 ATP/bit
(thermodynamic floor)
biological overhead
APGI bridge κ
≈ 100 ATP/bit
(network-state
bit-count convention)
whole-event thermodynamic inefficiency ≈ 1, 709 ×
the Landauer minimum (distinct from κ, the per-bit unit cost)
B
Per-ignition energy budget
1. ATP per spike
2.4e+09 ATP
(Attwell & Laughlin 2001;
scaled per Lennie 2003)
2. Spikes per ignition
1e+09 spikes
(~10^8 active neurons
x ~10 spikes / ~500 ms)
3. Total ATP per ignition
2.4e+18 ATP
(order-of-magnitude)
C
Bit-count convention determines κ
Percept-level coding
(~1-2 bits/spike)
~10^9-10^{10} bits
κ ≈ 10^8-10^9 ATP/bit
Network-state activation
patterns (~10^{12} synapses)
~2×10^{16} bits
κ ≈ 10^2 ATP/bit
Six orders of magnitude apart — the [10, 1,000] ATP/bit
falsification window is conditional on the network-state convention.